

Tree growth responses to extreme drought events are not well predicted by climate

Victor Van der Meersch¹, Benjamin I. Cook^{2,3},
Michael Betancourt⁴, E.M. Wolkovich¹

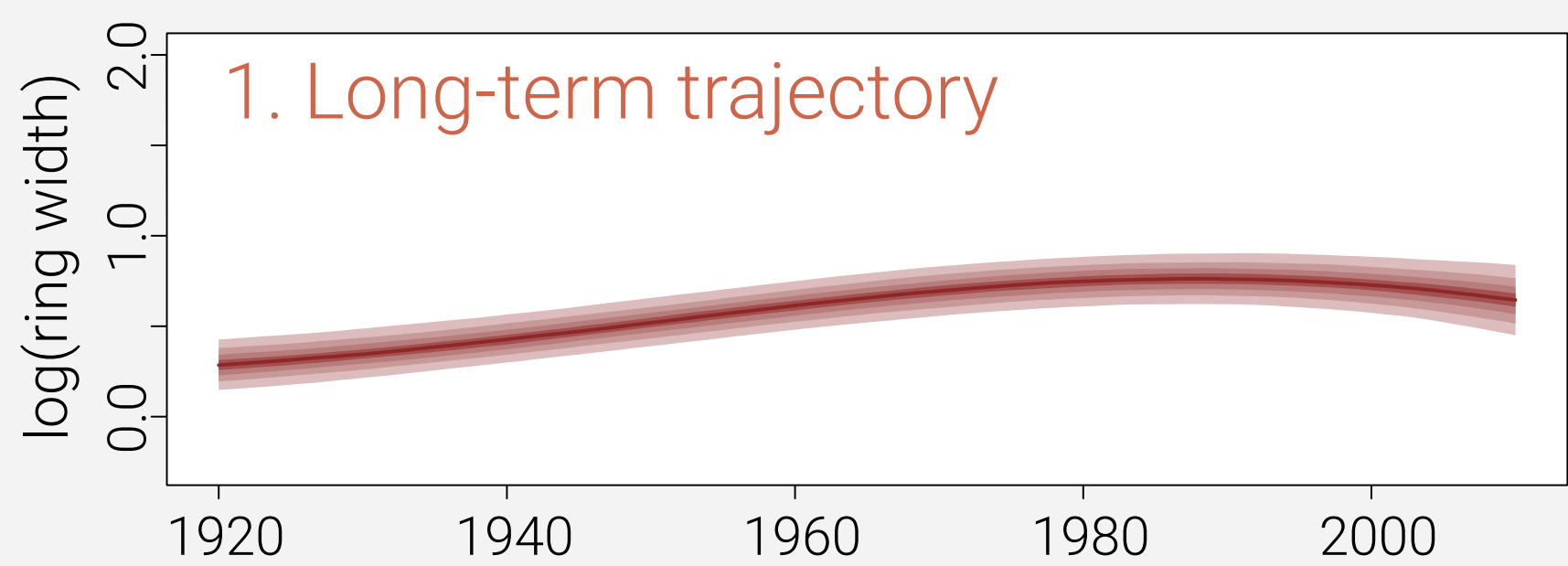
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⁴ Symplectomorphic LLC, New York, USA

OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture [1]
- Combined effects of these drivers on tree growth remain poorly understood at large scales [2,3]
- A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
- Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
- Current models likely **underestimate growth declines** during extreme events

1. MODELING GROWTH AT THE TREE LEVEL

- Tree-ring width time series are **one of the longest and most spatially extensive** records of forest response to climate
- However, tree rings are affected by **many drivers other than climate**: age-size effects, competition, biotic disturbance...
- Tree growth results from three major components that we jointly model:

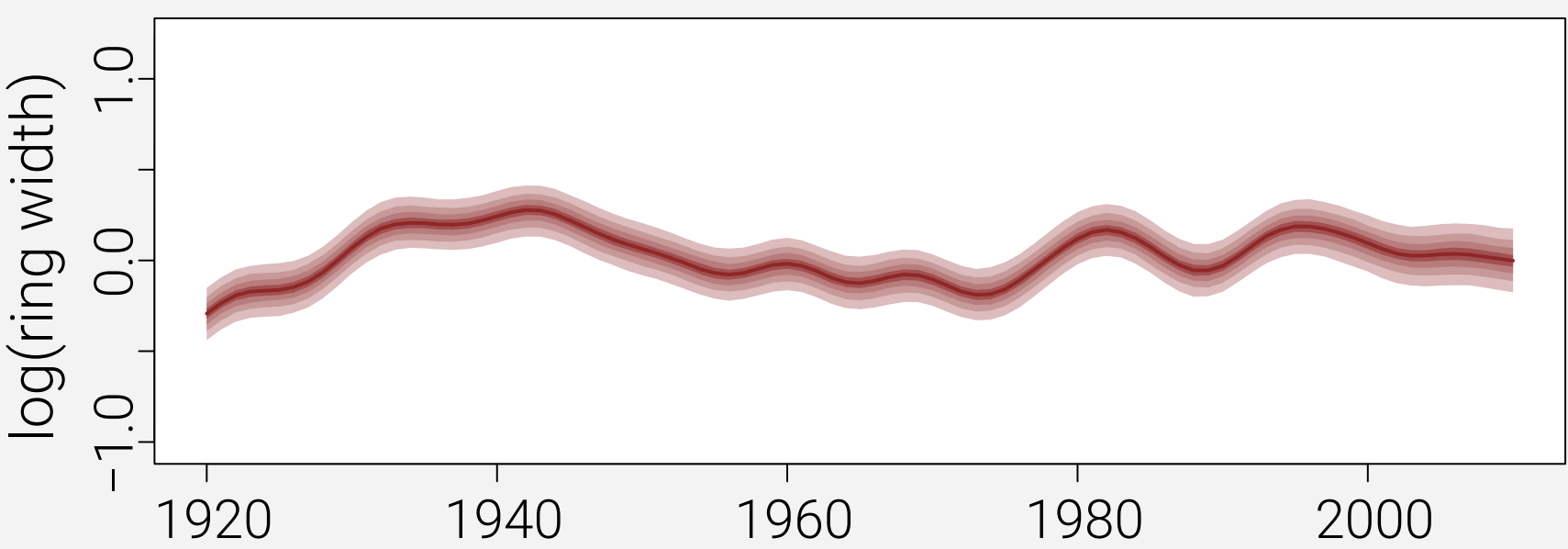


- A **long-term GP** for processes such as allometric constraints, for each tree:

$$\vec{f}_{\text{long}} \sim \text{multinormal}(\vec{0}, \Sigma_{\text{long}})$$

characterized by an exponentiated quadratic kernel with species-specific length scale ρ_{sp} and marginal variation γ_{sp}

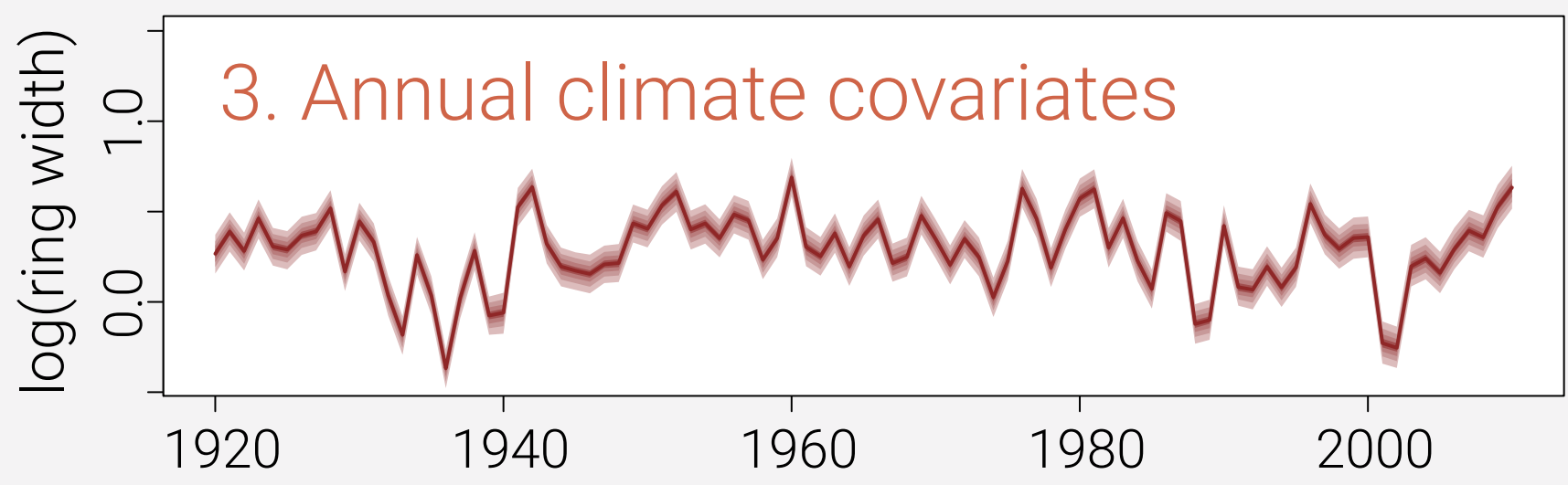
2. Mid- to short-term dynamics



- A **short- to mid-term GP** for processes such as neighbour competition dynamics:

$$\vec{f}_{\text{short}} \sim \text{multinormal}(\vec{0}, \Sigma_{\text{short}})$$

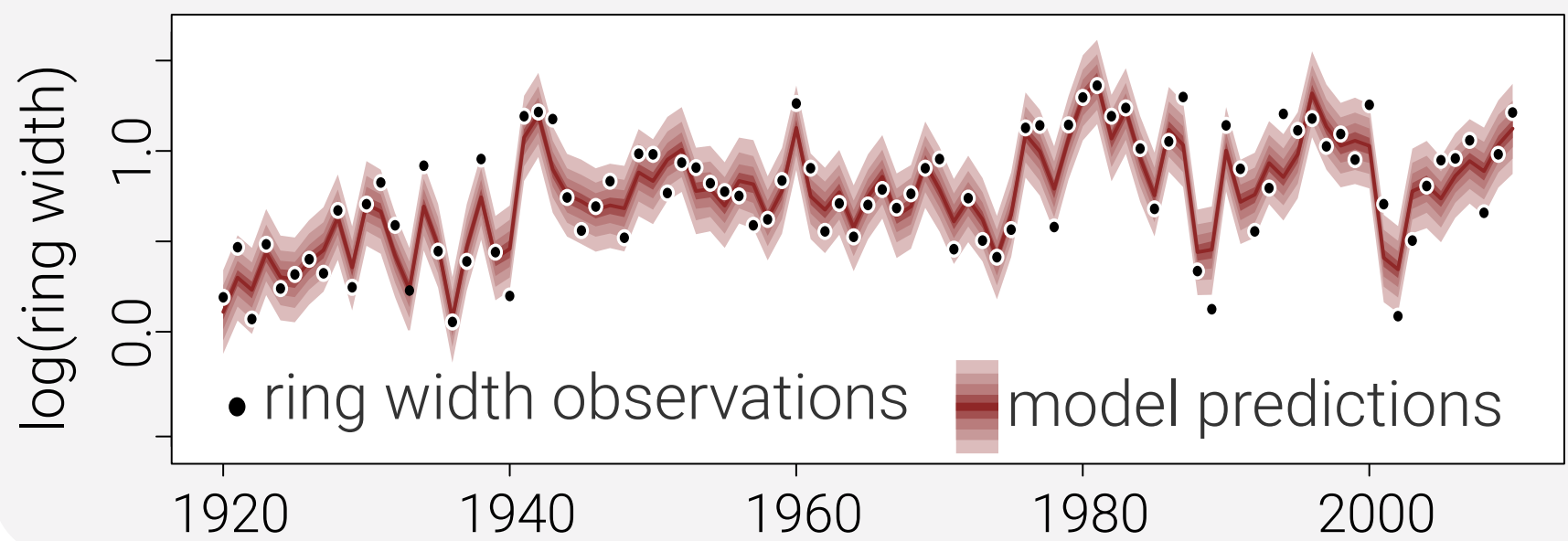
characterized by a second exponentiated quadratic kernel with length scale ρ_{short} and marginal variation γ_{short}



- A **linear regression** of 3 climate variables with species-specific slopes $\vec{\beta}_{\text{species}}$

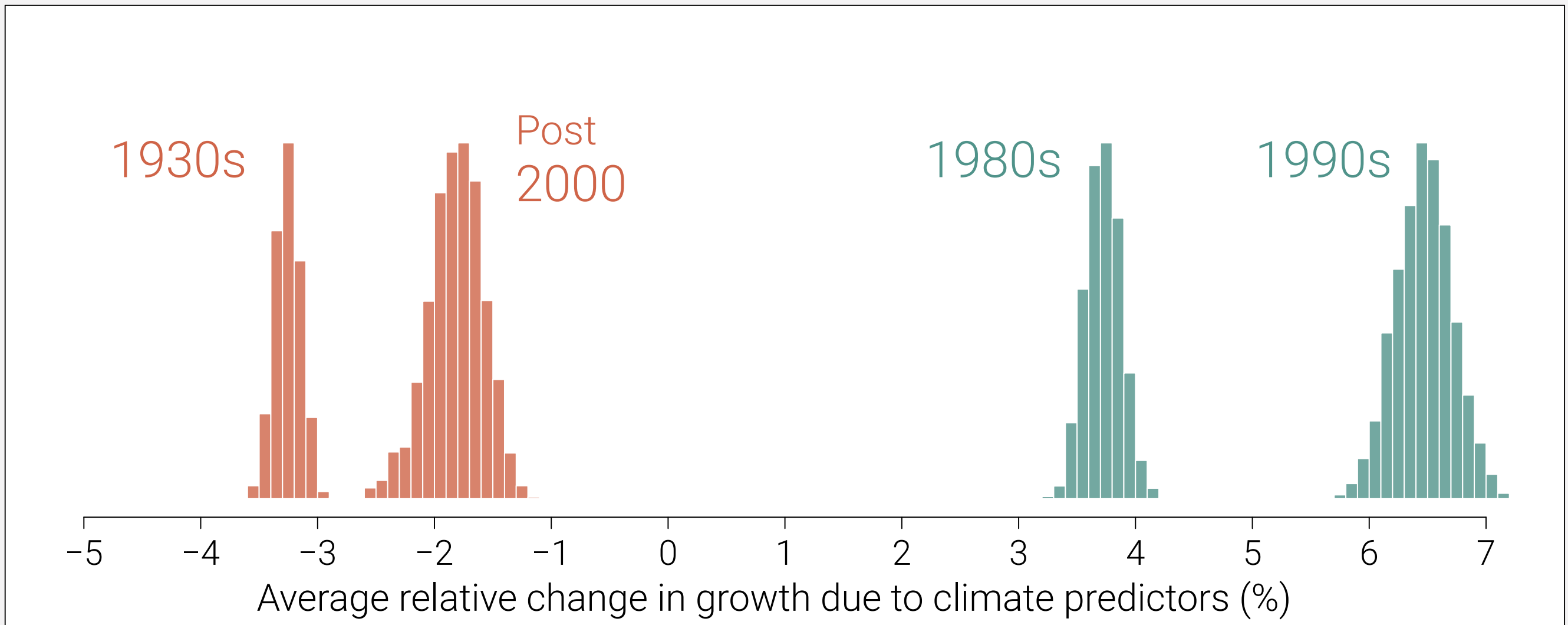
- All together, we have, for each observation of each tree:

$$\log(y_i) \sim \text{normal}(\alpha + f_{\text{long},i} + f_{\text{short},i} + \vec{\beta} \cdot \vec{X}_{\text{clim},i}, \sigma)$$



3. MAIN CLIMATIC DRIVERS OF GROWTH

Growing degree days, winter precipitation, and vapor pressure deficit lead on average to **±10% of changes in ring width**.

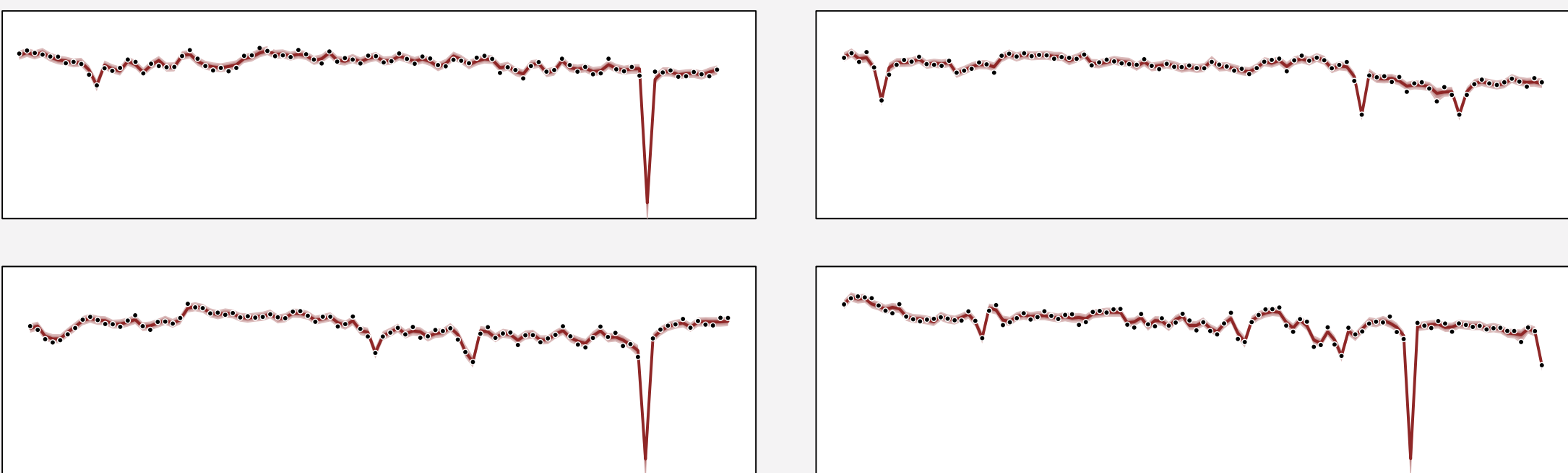
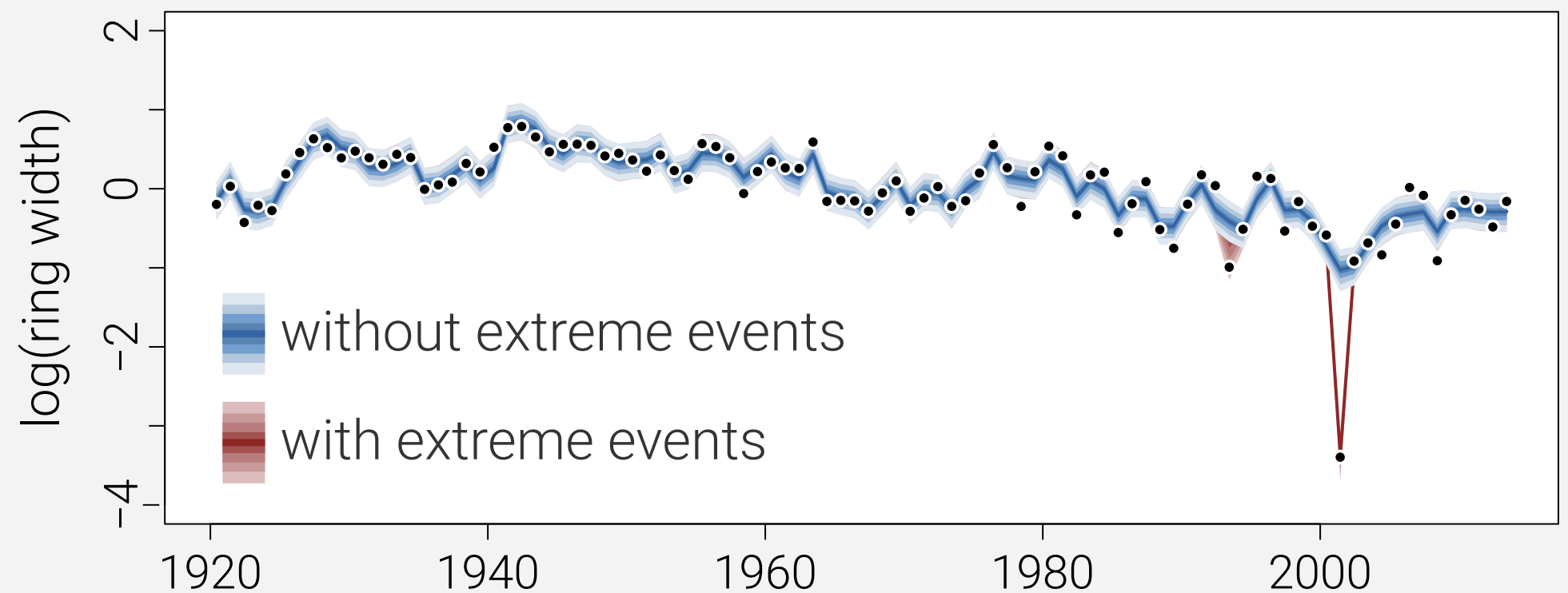


Our climate-growth component generally captures some known **wet periods** and some **less favorable periods** for growth.

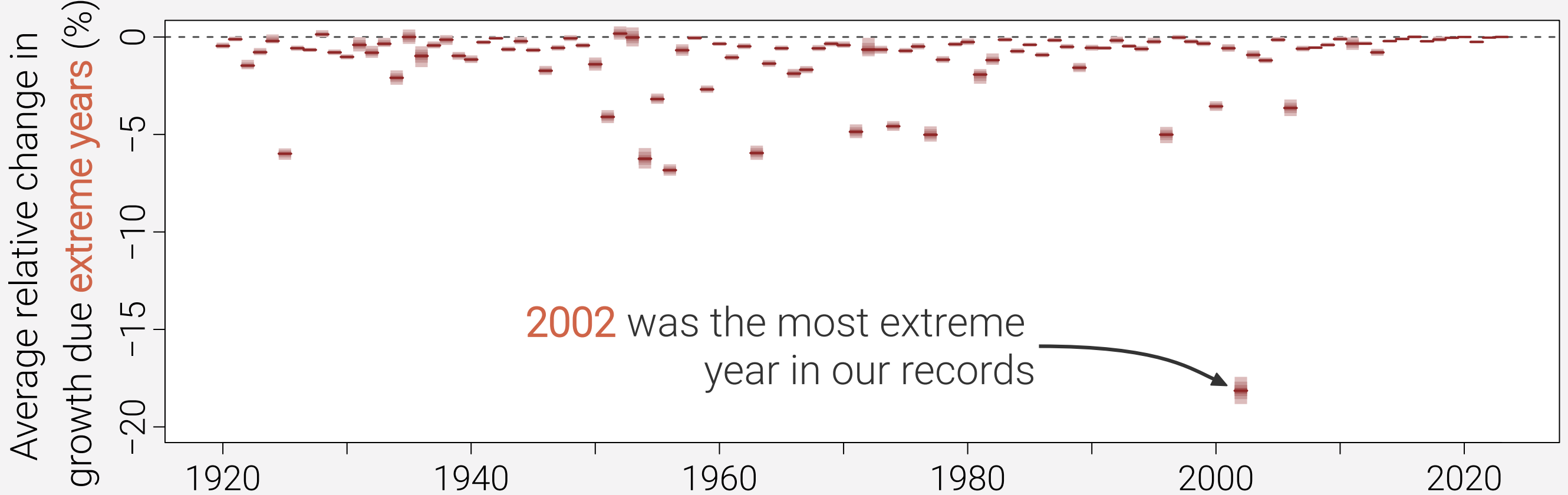
4. EXTREME GROWTH REDUCTIONS (SHOCKS)

But a tree can show **severe growth suppression**, far beyond what its climate response predicts.

When one tree experiences a shock, **neighboring trees within the same forest** tend to as well.



These shocks have major **consequences at the ecosystem scale**.



The synchrony of shocks reveals a shared, forest-level signal, consistent with a **common external stressor** (rather than individual chance).

Many of these shocks correspond to known major drought events, which **suppress growth by up to almost 20%** on top of the effects of climate predictors.

5. IMPLICATIONS FOR FORESTS AND FORECASTS

- Classical dendroclimatology extracts climate signals by averaging out individual variation (top-down), while here **we model individual tree growth** (bottom-up)
- Climatic drivers capture inter-annual to decadal variability, but **systematically miss extreme growth shocks** that are likely due to individual endogenous processes [5]
- Accounting for these **non-linear, individual-level responses is critical for accurate forecasts** of forest vulnerability and carbon storage under climate change [6]

References

- [1] Gebrechorkos et al. Nature (2025)
- [2] Babst et al. Sci Adv (2019)
- [3] Gao et al. Nat Ecol Evol (2022)
- [4] Melvin & Briffa. Dendrochronologia (2008)
- [5] Grossiord et al. New Phytologist (2020)
- [6] Anderegg et al. Nat Clim Change (2020)

Modeling the complexity of tree growth with hierarchical Gaussian processes and structured shocks

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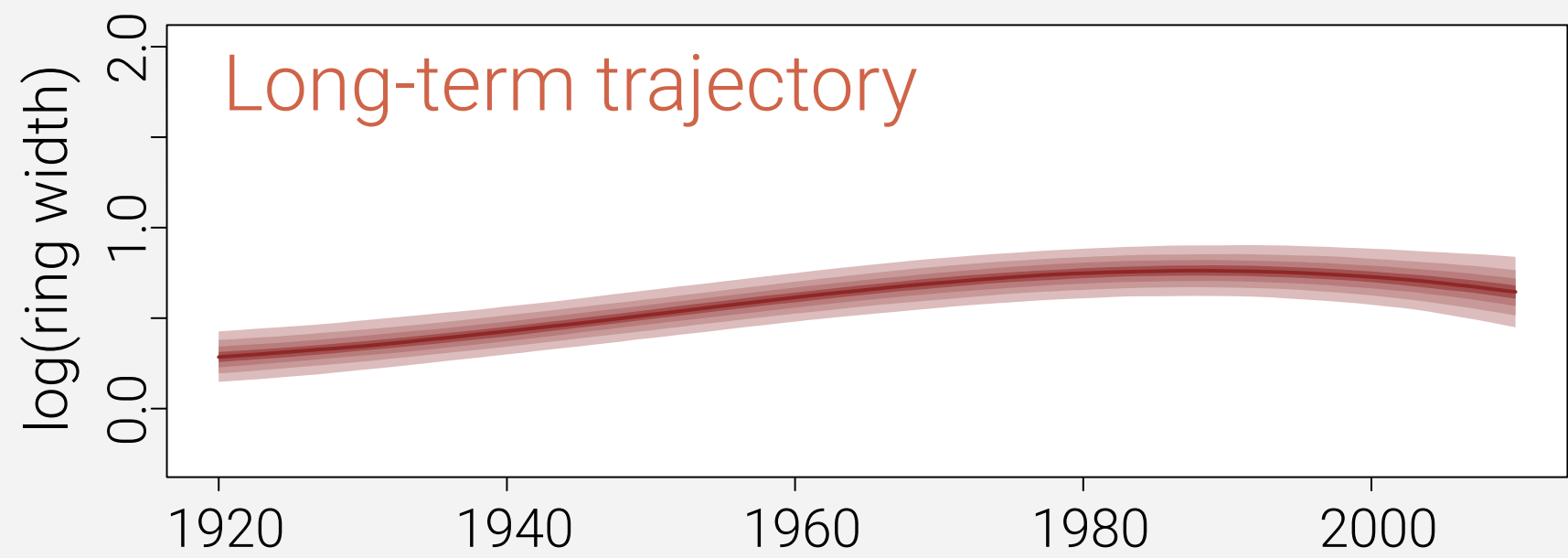
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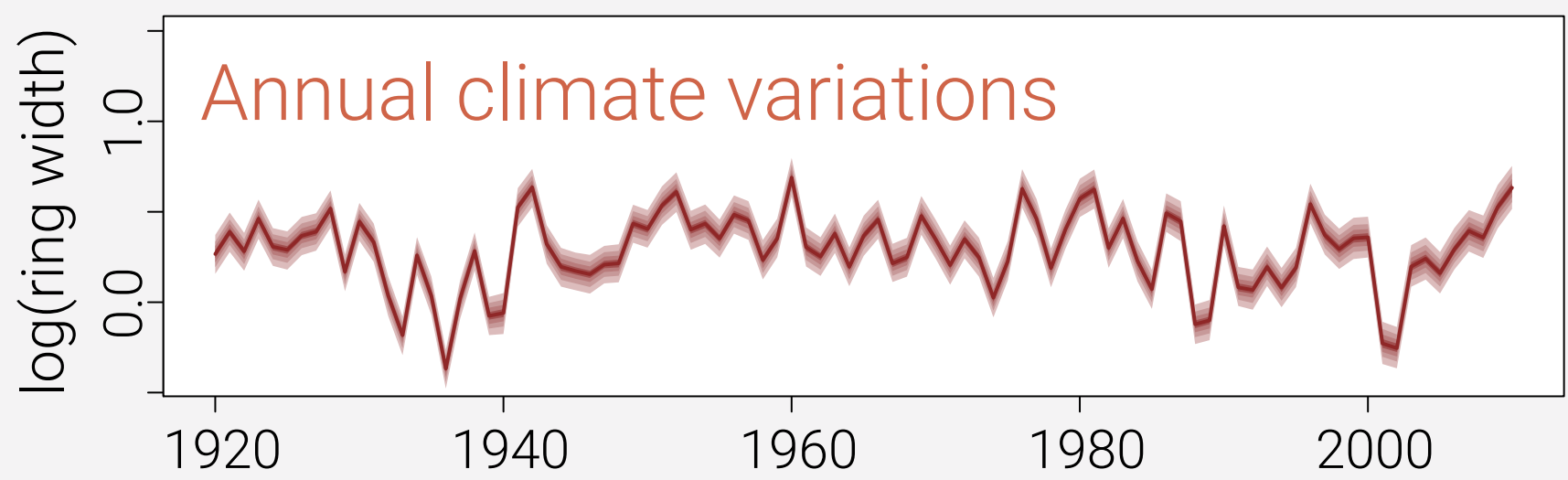
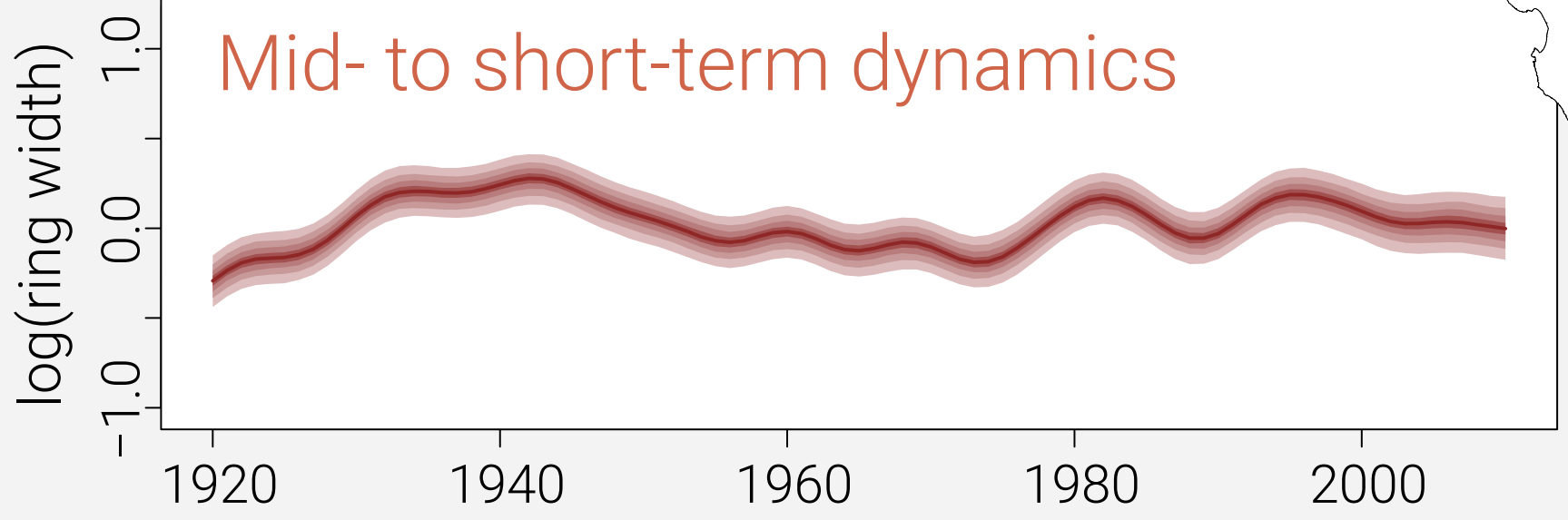
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- However, tree rings are affected by **many drivers other than climate**: age-size effects, competition, biotic disturbance...
- We model jointly **3 components**:



1. A **long-term GP** for processes such as allometric constraints:
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characterized by an Gaussian kernel with species-specific length scale ρ_{sp} and marginal variation γ_{sp}

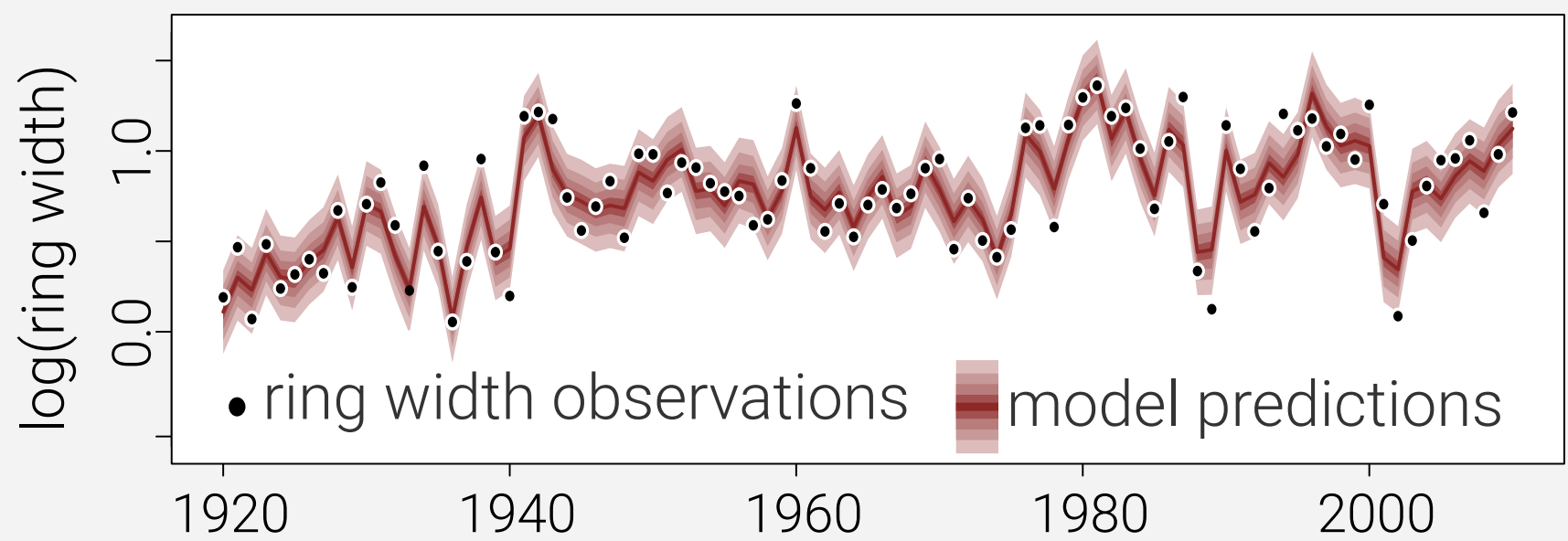
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3. A **linear regression** of 3 climate variables with species-specific slopes $\vec{\beta}_{\text{sp}}$

All together, we have, for each observation of each tree:

$$\log(y_i) \sim \text{normal}(\alpha + f_{\text{long},i} + f_{\text{short},i} + \vec{\beta}_{\text{sp}} \cdot \vec{X}_{\text{clim},i}, \sigma)$$



2. EXTREME GROWTH REDUCTIONS (SHOCKS)

- But a tree can show **severe growth suppression**, far beyond what its climate response $\vec{\beta}_{\text{sp}} \cdot \vec{X}_{\text{clim},i}$ predicts
- When one tree experiences a shock, **neighboring trees within the same forest** tend to as well, consistent with a **common external stressor** (rather than individual chance)
- We model this with **nested latent states**:

1. At the **forest level**, a stress can happen the year t with a probability that depends on the climate:

$$p(\text{forest-level stress at } t) = \text{logit}^{-1}(\alpha_{\text{forest}}^{\text{stress}} + \vec{\beta}_{\text{stress}} \cdot \vec{X}_{\text{clim},t})$$

2. Given that a stress happens, a **tree** might experience a shock with some probability that depends on the species:

$$p(\text{tree shock} \mid \text{forest-level stress at } t) = \text{logit}^{-1}(\alpha_{\text{forest}}^{\text{shock}} + \alpha_{\text{sp}}^{\text{shock}})$$

If a shock happens, this can lead to a growth depression of a species-specific amplitude τ_{sp} ... or even an entire shutdown

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